Tunisian Administrative Procedures Assistant

INSTALLATION AND TESTING GUIDE FOR PROFESSORS

Windows 11 | Telegram interface | PowerShell commands

Prepared 19 October 2026. Use this guide with the complete project archive or its approved repository URL. Replace <APPROVED_REPOSITORY_URL> before cloning; if using an archive, extract it and open PowerShell in its root folder.

Command convention: every command block is entered in Windows PowerShell. Commands beginning with `docker compose exec` execute the following program inside a container. Browser actions are described separately.

This guide initializes a **new local installation**, not production. Commands are PowerShell, from the repository root. The release worktree has no remote: replace the URL placeholder with the approved publication URL.

1. Prerequisites

Mandatory: Git; Python 3.11 with the py launcher; Docker Desktop in Linux-container mode with Compose v2; native Ollama. Allow enough disk/RAM/VRAM for models and containers. No universal hardware minimum has been measured; CPU inference is slower. Internet is needed for packages/models and Telegram. A Telegram account/token is optional until real bot activation.

```
git clone <APPROVED_REPOSITORY_URL> tunisian-admin-assistant
Set-Location tunisian-admin-assistant
py -3.11 --version
docker version
docker compose version
ollama --version
py -3.11 -m venv .venv
.venv\Scripts\python -m pip install --upgrade pip
.venv\Scripts\python -m pip install -e ".[dev]"
```

Virtual-environment activation is optional when using explicit executable paths.

2. Models

Run native Ollama once; do not launch a competing server.

```
ollama pull bge-m3
ollama pull mistral:latest
ollama pull translategemma:4b
ollama list
Invoke-RestMethod http://127.0.0.1:11434/api/tags
```

BGE-M3 produces 1024-dimensional embeddings; Mistral generates grounded answers; TranslateGemma translates checklist items FR-to-EN. Arabic/Derja also use official text and deterministic processing; universal translation quality is not claimed. Record model digests for exact reproduction. Containers use `host.docker.internal:11434`. The optional `ollama-docker` profile is unnecessary for this path.

3. Configuration

```
Copy-Item .env.example .env
notepad .env
```

Replace `N8N_ENCRYPTION_KEY` with a privately generated long random secret. Preserve it for the lifetime of the n8n volume. Never share tokens in chat or reports.

Variables	Purpose
`APP_ENV`, `LOG_LEVEL`	Runtime environment/logging
`DATA_DIR`, `SOURCES_REGISTRY`	Host data and registry paths; Compose supplies container paths
`QDRANT_URL`, `QDRANT_API_KEY`, `QDRANT_COLLECTION_PREFIX`	Host vector connection, optional key, namespace
`QDRANT_HOST_PORT`, `BACKEND_HOST_PORT`, `N8N_HOST_PORT`	Loopback ports: 6343, 8000, 5678
`OLLAMA_URL`, `OLLAMA_URL_FOR_CONTAINERS`	Host and container model-server URLs
`EMBEDDING_MODEL`, `EMBEDDING_DIM`, `LLM_MODEL`	Embedding and generation configuration
`TRANSLATION_MODEL`, `LLM_TIMEOUT_SECONDS`, `INTENT_LLM_FALLBACK`, `RETRIEVAL_MIN_SCORE`	Local model/routing overrides; Compose uses code defaults
`HTTP_USER_AGENT`, `HTTP_TIMEOUT_SECONDS`, `MAX_DOWNLOAD_BYTES`, `DEFAULT_CRAWL_DELAY_SECONDS`	Source acquisition settings
`MAX_UPLOAD_BYTES`	Upload limit, default 10 MiB
`N8N_ENCRYPTION_KEY`, `GENERIC_TIMEZONE`	Required n8n secret and timezone
`TELEGRAM_BOT_TOKEN`, `ASSISTANT_WEBHOOK_URL`	Optional bot token and host-run bridge destination
`ADMIN_API_KEY`	Enables privileged endpoints; empty disables them
`AGENT_API_KEY`	Optional backend authentication; n8n must send matching headers
`ASSISTANT_WEBHOOK_KEY`	Optional bridge header; configure n8n webhook authentication first

Optional commented settings cover retrieval mode/diversity, evidence checks, OCR languages, request size, prompts and TLS paths. Exact defaults: `backend/settings.py`. **Compose forwards only variables listed in its service environment sections.** An unforwarded `.env` value does not override the container; use a reviewed Compose override if necessary.

Initially leave agent/webhook keys empty unless matching n8n authentication is configured. Services bind to loopback; do not expose unauthenticated endpoints publicly. `.env` is excluded from Git and Docker contexts.

4. Offline corpus reconstruction

```
.venv\Scripts\python scripts/build_processed_corpus.py
```

This reads tracked official originals and writes ignored `data/processed/` extraction/chunk files. No downloading, embedding or Qdrant writes occur. Expected: **14 documents, 286 structural chunks, 186 recursive chunks**. One image-only decree is skipped. Historical production has **12 indexed documents, 272/174 points**; CIN/passport additions explain the fresh-index difference. Fifteen manifest records are not fifteen indexed documents.

5. Start the new stack

Verify ports are free. Compose has a fixed project name and bridge container name: do not run this alongside production without a reviewed isolation override. A different checkout alone does not isolate Docker resources.

```
docker compose config --quiet
docker compose up -d --build
docker compose ps
Invoke-RestMethod http://127.0.0.1:8000/health
Invoke-RestMethod http://127.0.0.1:5678/healthz
docker compose exec backend tesseract --list-langs
```

The backend image installs Tesseract with French/Arabic packs. Native OCR tests additionally need Tesseract on PATH with `fra` and `ara` data. The public Sectigo intermediate in `config/certs/` repairs an incomplete official-server TLS chain; it is not a secret. Never disable TLS verification or bypass source policies.

6. Initial Qdrant indexing - NEW EMPTY INSTALLATION ONLY

`/index/run` **recreates collections**. First inspect the target:

```
Invoke-RestMethod http://127.0.0.1:6343/collections
```

If valuable collections exist, STOP. On a new empty instance only, privately set a temporary `ADMIN_API_KEY` in `.env`, then load it into the new backend:

```
docker compose up -d --no-deps backend
$adminSecret = Read-Host 'Temporary admin key' -AsSecureString
$adminPointer = [Runtime.InteropServices.Marshal]::SecureStringToBSTR($adminSecret)
try {
    $adminPlain = [Runtime.InteropServices.Marshal]::PtrToStringBSTR($adminPointer)
    $indexResult = Invoke-RestMethod -Method Post -Uri http://127.0.0.1:8000/index/run -Headers `
    @{'X-Admin-Key'=$adminPlain} -TimeoutSec 1800
    $indexResult.collection_points
} finally {
    [Runtime.InteropServices.Marshal]::ZeroFreeBSTR($adminPointer)
    Remove-Variable adminPlain,adminSecret -ErrorAction SilentlyContinue
}
```

Expect 286/186 points for this snapshot. Remove the temporary admin key from `.env` and recreate only this new backend again. No ingestion/download is necessary. Workflow 01 must stay inactive and unexecuted.

7. Import and publish workflows - fresh n8n database only

Open `http://localhost:5678`, create the local owner account privately, then initialize:

```
docker compose exec n8n n8n import:workflow --separate --input=/workflows
docker compose exec -e N8N_RUNNERS_BROKER_PORT=5690 n8n n8n publish:workflow --id=TnaaRetrieveApi1
docker compose exec -e N8N_RUNNERS_BROKER_PORT=5690 n8n n8n publish:workflow --id=TnaaAssistant01
docker compose restart n8n
```

The restart loads publication/webhook registration in this new installation. These initial-import commands are NOT the safe update procedure for existing live workflows. Existing synchronization needs backups, authenticated API updates and version/state verification.

Verify in the native editor:

• `TnaaIngestIndex1`: inactive, 5 executable nodes and 6 notes; never execute during installation.

• `TnaaRetrieveApi1`: published, 3 executable nodes and 4 notes; `/webhook/retrieve`.

• `TnaaAssistant01`: published, 17 executable nodes and 10 notes; `/webhook/assistant`.

HTTP nodes use `http://backend:8000` within Docker. Host clients use loopback ports. No real credential is embedded in the exports; API/owner credentials differ from webhook authentication.

8. HTTP checks and optional Telegram activation

```
$question = @{text='Quels documents pour renouveler mon passeport ?';language='fr'} | `
    ConvertTo-Json
Invoke-RestMethod -Method Post -Uri http://127.0.0.1:5678/webhook/assistant -ContentType `
    'application/json; charset=utf-8' -Body ([Text.Encoding]::UTF8.GetBytes($question)) `
    -TimeoutSec 600
```

A clarification about residence may be appropriate. Check supporting sources, not one fixed wording. Upload JSON has `file` with `filename`, `mime_type`, `data_b64`, plus `language`. Dossier requests add `mode: "dossier"`; ordinary Form Check omits mode.

For a real bot, contact official verified `@BotFather`, send `/newbot`, choose a display name and available username ending in `bot`. Perform login/verification yourself. Enter the issued token only in private `.env`. Use a dedicated test bot, never a token already polled by production.

```
docker compose --profile telegram up -d --no-deps --build telegram-bridge
docker compose ps
```

Without a token the bridge waits in `LIVE_CONNECTIVITY_PENDING`. Health does not prove delivery. Manually test `/start`, `/help`, `/language`, contextual questions, `/form_check`, `/dossier_check`, PDF/image uploads, refusal and `/cancel`. Do not share unsanitized logs. See Telegram activation.

9. Automated and end-to-end testing

```
.venv\Scripts\python -m pytest -m "not integration" -q
# Dedicated initialized test installation using the expected ports only:
.venv\Scripts\python -m pytest -m integration -q
```

Some integration modules/scripts hard-code localhost:5678/8000/6343. Environment overrides alone do not isolate them. Never run them beside production without inspecting destinations. Native OCR tests may skip without Tesseract. End-to-end acceptance includes the HTTP check and actual manual Telegram exchanges; do not claim the latter from unit tests.

10. Troubleshooting and safe shutdown

Problem	Resolution
Missing encryption key	Set it privately; preserve keys already associated with a volume
Port/container-name conflict	Stop setup; do not remove unknown containers
Ollama unavailable/model missing	Check host server, container reachability and `ollama list`
Missing extraction/checklists	Run the offline builder in this checkout
Empty Qdrant	Confirm new empty instance before indexing
Webhook 404	Verify publication, production webhook path and registration after startup
HTTP 503	Check upstream availability; some workflow errors deliberately map to 503
Index HTTP 403	Admin endpoint is disabled without a configured key
OCR refusal	Use a clearer supported test form; do not lower thresholds blindly
Telegram 409	Another poller uses the token; stop the test poller, not production

```
docker compose --profile telegram stop
docker compose --profile telegram start
```

Verify project ownership before lifecycle commands. Do not use `down -v`, volume prune or collection deletion for routine shutdown. Preserve private configuration and encryption keys; back up volumes before upgrades. Keep uploads, backups and credential exports out of Git.